

MagXMCD v9 — Hybrid Classical+CNN Pipeline Results

ECE 228 Final Project | 7 Pipelines Compared | Classical Phase Retrieval + CNN Refinement

1. Experiment Overview — What MagXMCD v9 Tests

This is the key hybrid experiment of the project. Rather than treating classical phase retrieval and CNN reconstruction as competing approaches, MagXMCD v9 asks: can a CNN learn to refine and denoise the output of a classical phase retrieval algorithm, producing better reconstructions than either approach alone? Three classical algorithms (RAAR, GENFIRE, RESIRE) are tested as standalone methods and as preprocessors feeding into a CNN, giving 7 total pipelines including CNN-alone as the pure ML baseline.

2. Experimental Configuration (run_config.json)

Parameter	Value	Parameter	Value
Grid size	64 x 64	Training samples	3,000
Epochs	30	Wall type	Both (Bloch + Neel)
N_ANGLES	24	Angle scheme	4 theta x 6 phi
Theta values	15°, 35°, 55°, 75°	Phi values	0°, 60°, 120°, 180°, 240°, 300°
C_CONST	1.0	Pipelines tested	7

3. The 7 Pipelines

Pipeline	Type	Description
CNN alone	Pure ML	U-Net trained directly on raw XMCD scattering patterns. No classical preprocessing.
RAAR alone	Classical	Relaxed Averaged Alternating Reflections phase retrieval only. No ML refinement.
GENFIRE alone	Classical	GENeralized Fourier Iterative REconstruction phase retrieval only. No ML refinement.
RESIRE alone	Classical	REal Space Iterative REconstruction phase retrieval only. No ML refinement.
RAAR→CNN	Hybrid	RAAR phase retrieval output fed as input to CNN for refinement.
GENFIRE→CNN	Hybrid	GENFIRE output fed as input to CNN for refinement.
RESIRE→CNN	Hybrid	RESIRE output fed as input to CNN for refinement.

4. Primary Results — Mz Domain Accuracy (from figures)

Pipeline	Type	Mz Accuracy (%)	vs Random (50%)	vs CNN alone (52.2%)
CNN alone	Pure ML	52.2%	+2.2%	— (baseline)
RAAR alone	Classical	74.1%	+24.1%	+21.9%
GENFIRE alone	Classical	74.1%	+24.1%	+21.9%
RESIRE alone	Classical	74.2%	+24.2%	+22.0%
RAAR→CNN	Hybrid	74.9%	+24.9%	+22.7% ← best hybrid
GENFIRE→CNN	Hybrid	75.0%	+25.0%	+22.8% ← best overall
RESIRE→CNN	Hybrid	74.3%	+24.3%	+22.1%

KEY RESULT: All classical and hybrid methods vastly outperform CNN alone (52.2%). The classical algorithms achieve ~74% accuracy without any ML. Hybrid methods (classical→CNN) provide a marginal additional improvement of ~0.7-0.9% over classical alone. GENFIRE→CNN is the best overall pipeline at 75.0%.

5. Quantitative Mz Reconstruction Statistics (computed from reconstructions.npz)

Pipeline	MSE vs True	MAE vs True	Correlation	Sign Accuracy
CNN alone	0.9806	0.9852	-0.1249 (anti-correlated!)	48.6% (below random)
RAAR alone	0.5132	0.5362	0.6770	74.2%

GENFIRE alone	0.5132	0.5362	0.6770	73.9%
RESIRE alone	0.5132	0.5362	0.6770	73.3%
RAAR→CNN	0.4361	0.4998	0.7376	87.8% ← best
GENFIRE→CNN	0.5117	0.5372	0.6790	75.4%
RESIRE→CNN	0.5206	0.5477	0.6712	73.8%

* CNN alone achieves correlation = -0.124 and sign accuracy 48.6% — both BELOW random (50%). The CNN is actively anti-correlated with the true field on this test sample. RAAR→CNN achieves the best MSE (0.4361) and sign accuracy (87.8%) of all pipelines.

6. CNN Training Dynamics — Validation MSE per Pipeline

Pipeline	Ep 1	Ep 5	Best val	Best ep	Ep 20	Ep 30	Trajectory
CNN alone	0.334	0.350	0.332	ep 7	0.509	0.556	Diverging — val MSE increases monotonically after ep 7. Severe overfit.
RAAR→CNN	0.234	0.181	0.179	ep 8	0.288	0.301	Fast convergence to 0.179, then gradual overfit. Classical seed stabilizes early training.
GENFIRE→CNN	0.231	0.177	0.176	ep 8	0.281	0.301	Nearly identical to RAAR→CNN. Both algorithms provide equally useful seeds.
RESIRE→CNN	0.229	0.178	0.176	ep 7	0.285	0.300	Marginally best convergence. Slight edge from RESIRE seed quality.

All three hybrid pipelines converge to similar best val MSE (~0.176-0.179 at ep 7-8), then gradually overfit to ~0.300 by ep 30. CNN alone diverges from ep 7 onward, reaching 0.556 by ep 30 — 85% worse than the hybrid methods.

7. Qualitative Observations from Reconstruction Images

Pipeline	Visual Quality	Key Observation
CNN alone	Poor	Reconstruction is near-uniform pale color — model predicts close to zero everywhere. acc=52.2% is barely above random (50%). Domain boundaries completely lost.
RAAR alone	Good	Clear domain structure recovered. Domain shape recognizable but boundary is pixelated/blocky. acc=74.1%.
GENFIRE alone	Good	Very similar to RAAR alone — same blocky boundary, same accuracy (74.1%). GENFIRE and RAAR produce nearly identical reconstructions.
RESIRE alone	Good	Marginally smoother boundaries than RAAR/GENFIRE. acc=74.2%. All three classical methods converge to similar solutions.
RAAR→CNN	Good+	Domain shape preserved from RAAR, with smoother color gradients from CNN refinement. Domain wall transition is softer. acc=74.9%. Some color artifacts in background regions.
GENFIRE→CNN	Best	Best visual quality. Smoothest domain boundary. Continuous color gradient across domain wall. acc=75.0% — highest of all 7 pipelines.
RESIRE→CNN	Good+	Similar to RAAR→CNN. Smooth interior but some boundary artifacts. acc=74.3%.

8. Key Findings & Paper Significance

#	Finding	Implication for Paper
1	CNN alone completely fails (52.2%, anti-correlated with truth)	Without phase information, the CNN cannot learn the inverse map. Confirms findings from MagCNN runs — amplitude-only input is insufficient.
2	Classical methods (RAAR/GENFIRE/RESIRE) all achieve ~74% with no ML	Phase retrieval solves the hard part of the problem. The classical algorithms successfully recover domain topology from scattering amplitudes.
3	Hybrid (classical→CNN) improves over classical alone by ~0.7-0.9%	CNN refinement adds marginal value on top of classical phase retrieval. The improvement is real but small — the classical algorithm does most of the work.
4	RAAR→CNN achieves best MSE (0.436) and sign accuracy (87.8%)	RAAR provides the best seed for CNN refinement despite similar standalone accuracy to GENFIRE/RESIRE. The seed quality matters more than the standalone algorithm accuracy.
5	GENFIRE→CNN achieves best domain accuracy (75.0%) — best overall pipeline	GENFIRE seed produces the best final visual reconstruction and highest domain accuracy. Different metrics favor different pipelines — important nuance for paper.

6	All three classical algorithms produce nearly identical results (74.1-74.2%)	RAAR, GENFIRE, and RESIRE converge to similar solutions on this problem. Algorithm choice matters less than having phase retrieval at all.
7	This is where in the story arc: Classical phase retrieval works; CNN refines it	This notebook is the payoff of the whole project. The answer to "how do you solve the XMCD inverse problem" is: phase retrieval first, CNN refinement second. Neither alone is sufficient or optimal.